

World Models for Physical AI: Learning to Predict, Plan, and Act in the Real World

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World models provide a means of representing how physical environments evolve and how an agent’s actions may change them. Their relevance to Physical AI extends across predictive representation learning, simulation, planning, policy training, and execution monitoring. These roles place different demands on what a model predicts, how actions condition its predictions, and how the predictions are used. This survey examines world models through these three questions, connecting model-based control with visual and latent prediction, structured physical representations, multimodal interaction data, and robot policies. We organize the discussion into foundations, data and model design, integration with decision making, and evaluation and deployment. We further examine reasoning-policy hybrids as an emerging interface between general-purpose reasoning and robot control. The organizing objective is to connect each system-level claim to evidence appropriate to its role, distinguishing predictive quality, decision utility, and operational reliability.

CCS Concepts: • **Computing methodologies** → **Robotics**; *Reinforcement learning*.

Additional Key Words and Phrases: world models, Physical AI, robot learning, predictive representations, vision-language-action models, model-based control

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1 Foundations and Scope

1.1 Introduction

Physical AI concerns agents whose decisions change a physical environment. For a robot, choosing an action requires more than identifying a goal: the system must also contend with what is currently observable, how its body can move, and how objects may respond. A world model makes some of these consequences available as predictions. The central question for this survey is how such predictions contribute to learning and acting under the constraints of physical interaction.

Predictive models can support control through different computational routes. In *World Models*, learned visual and temporal representations support policy learning, including training inside a learned environment [10]. PlaNet instead learns latent dynamics from images and uses online planning to select actions [12]. These examples motivate a distinction between using prediction to construct a policy and querying predictions while choosing an action. Their experimental settings establish particular uses of learned dynamics rather than a general guarantee of real-world competence.

Robot policies offer a complementary interface: they map observations and instructions to actions. OpenVLA, for example, adapts a vision-language backbone using robot demonstrations to produce visuomotor behavior [21]. This interface does not by itself expose a future-state predictor that can be queried under alternative actions. The distinction is functional: a policy may contain useful knowledge of dynamics, while an explicit prediction interface provides an additional object that can be inspected, optimized against, or compared with subsequent observations.

The relationship between representation learning and action-conditioned prediction also deserves separate treatment. V-JEPA 2 combines action-free video pretraining with a subsequent action-conditioned model, V-JEPA 2-AC, for robotic planning [3]. The stages serve different purposes and should not be assigned the same control interpretation solely because they share a representation. This motivates recording both the supervision that creates a representation and the interface through which a controller later uses it.

We organize the survey around three questions: *what is predicted, how actions enter the model, and where prediction is used*. These are the survey’s comparison axes. Together they connect data and model design to operational roles without requiring all systems to share one architecture or output space. A video generator, a latent predictor, and a structured dynamics model may address different parts of this design space; their comparison should identify the relevant state variables, time horizon, and control interface before assessing performance.

The same role-based approach guides evaluation. Evidence that a model predicts observations accurately supports a different claim from evidence that its predictions improve action selection. Likewise, a closed-loop task result must be interpreted with its task distribution, controller, inference budget, and failure conditions. Our synthesis therefore records predictive performance, downstream use, and deployment conditions separately. This is an analytical choice of the survey, not a universal ranking of papers or a requirement that every supporting component demonstrate a complete robot system.

The article follows four parts. Section 1 establishes the scope and comparison axes. Section 2 examines the data, representations, and modeling choices that determine which consequences can be learned. Section 3 studies how prediction interacts with policies, planning, memory, and recovery. Section 4 organizes evaluation, reliability, safety, and open problems around the claims that a deployed system must substantiate. A companion literature catalog retains the seven collection categories used by the project and links them to the writing process.

1.2 Working Definitions and Inclusion Boundaries

We use *world model* for a learned model that predicts aspects of an environment’s evolution from available information. For an action-conditioned world model, the inputs additionally specify actions or other interventions. The predicted variables may be observations, latent states, object relations, physical quantities, or task-relevant consequences. This working definition accommodates models learned from passive observations as well as interaction data; it does not assume that every predictive representation already provides an executable robot-action interface.

For analysis, let a history summarize the agent’s observations and past actions. A policy maps this information and a goal to an action or action sequence. An action-conditioned world model maps the history and a proposed action sequence to predicted consequences. A controller turns the selected command into realized motion. Separating these interfaces lets us ask whether a failure originates in prediction, action selection, or execution, even when some interfaces are implemented inside a shared network.

The literature corpus includes both direct methods and supporting work. Learned simulators, predictive models used for planning, and prediction-guided policy learning provide direct cases. Datasets, analytical simulators, representation encoders, sensors, and policy baselines provide supporting context according to their demonstrated roles. Their inclusion does not imply that each item is itself a learned world model. In particular, the presence of a reasoning model in a control loop is insufficient to identify an explicit future-prediction mechanism.

Where does prediction enter the control loop?

Three functional interfaces for organizing the survey

A Direct action generation



B Policy proposals reviewed by a reasoner



C Action selection through explicit consequence prediction



Comparison axes Predicted variable Action interface Operational use

Conceptual interfaces. Reasoning alone does not establish an explicit future-state model.

Fig. 1. Interfaces linking observation to physical action. The schematic distinguishes direct action generation, policy proposals reviewed by a reasoning model, and candidate actions evaluated through an explicit consequence predictor. Arrows denote interface-level information flow, not the internal architecture of any one model. Observation feedback closes each loop.

1.3 From Learned Dynamics to Predictive Robot Systems

World-model research comprises interacting directions rather than a single replacement sequence. One direction learns dynamics for online decisions. PETS uses probabilistic ensembles and trajectory sampling to propagate uncertainty when evaluating actions [6]; PlaNet brings planning into a learned latent space [12]. Another direction uses predicted experience to train a policy. Dreamer optimizes behavior through imagined latent trajectories [11], whereas MBPO augments policy learning with short model rollouts initialized from collected experience [16]. The distinction concerns when the model is queried, not whether the final policy is a neural network.

Physical deployment adds sensing, actuation, and interaction costs to this design space. DayDreamer applies world-model learning directly to physical platforms for manipulation, locomotion, and navigation [39]. Foundation-model approaches enter from another direction: they acquire broad visual or semantic priors before adaptation to an action interface. Recent systems such as DreamZero couple future video with action generation [41]. These developments make it useful to retain both the classical model-use distinction and the newer questions of pretraining, embodiment, and execution.

With observation–action history h_t , goal g , and candidate action sequence $a_{t:t+H-1}$, a predictive model can be written as

$$p_{\theta}(y_{t+1:t+H} \mid h_t, a_{t:t+H-1}, g), \quad (1)$$

Table 1. Positioning against selected 2026 surveys. Stated scope is compared, not completeness.

Survey	Main organizing emphasis	Connection to this review
Kirchner et al. [22]	Physical AI: uncertainty, structure, decision coupling, and closed-loop control	Closest scope; establishes that control, uncertainty, and safety alone are not differentiating claims.
Hou et al. [14]	Robot learning: policy coupling, simulation, and predictive models	Broad method map for tracing primary papers.
Lu et al. [27]	World-action modeling, architecture, and control applications	Predictive and executable interfaces.
Jena et al. [17]	Benchmark protocols and predictive embodied evaluation	Prediction tests versus decision tests.
This review	Prediction target, action interface, operational use, and reliability	Connects design choices to evidence for physical use.

where y may represent images, latent features, geometry, or other task-relevant variables. A policy instead produces $\pi_\phi(a_{t:t+H-1} \mid h_t, g)$. These expressions describe interfaces rather than a required architecture. A joint model may implement both, while a model trained on passive video may omit the action argument. The practical question is which interface is trained, queried, and tested.

1.4 Relation to Recent Surveys and Review Scope

Recent surveys already cover substantial parts of this landscape. Kirchner et al. [22] provide a control-centric review of world models for Physical AI, with explicit attention to uncertainty, structural priors, and closed-loop reliability. Hou et al. [14] organize robot world models around policy coupling, learned simulation, and visual predictive modeling. Lu et al. [27] emphasize world-action architectures and control applications. Jena et al. [17] focus on evaluation benchmarks and the comparison between prediction and executed behavior. Table 1 summarizes their stated emphases without interpreting an omitted entry as proof that a topic is absent from a paper.

Our working emphasis is the traceable connection between a predicted quantity, the interface that uses it, and the evidence for physical interaction. This perspective accommodates world models, their training resources, and complementary policy or safety components without treating them as identical objects. The seven collection categories supply literature to four narrative parts: foundations establish interfaces; data and representations determine what can be predicted; integration explains how predictions affect behavior; evaluation examines whether that use is reliable.

The present manuscript is a first-pass narrative synthesis, not a completed systematic review. Its companion catalog contains 929 category records with cross-category duplication; this is neither a unique-study count nor a verified-paper count. Primary sources support method descriptions; surveys guide discovery and positioning. The selection log will be expanded alongside the manuscript, with formal publications preferred when available and preprints retained for explicitly identified recent developments.

2 Data, Representations, and Predictive Models

2.1 Data: From Observed Motion to Executed Actions

Training data determine which relationships a world model can learn directly and which require additional alignment. Passive video provides observations of interaction but does not necessarily contain the robot commands that produced it. Robot trajectories can pair images with commands and proprioception, while simulator rollouts can expose additional

Table 2. Example resources organized by supervision and use. Collection corpora, learned simulators, and task benchmarks remain distinct.

Resource / example	Learning signal	Role in a WM system	Extraction priority
Passive video; V-JEPA 2 [3]	Temporal visual observations	Representation before action adaptation	Which action labels are absent?
DROID [18]	Real manipulation demonstrations	Observation–action learning	Timing, sensors, and scene splits.
Open X-Embodiment [30]	Heterogeneous robot trajectories	Shared training across sources	Body, action units, and normalization.
UniSim [40]	Learned visual transitions	Interactive synthetic experience	Supported controls and rollout validity.
LIBERO / CALVIN [24, 29]	Tasks with interactive simulation	Downstream policy evaluation	Split, success test, and reset rules.

state variables. These resources provide different supervision; volume alone does not establish suitability for a particular control interface.

DROID supplies distributed real-robot manipulation demonstrations [18], while Open X-Embodiment aggregates robot datasets and studies cross-embodiment policy learning [30]. They illustrate diversity across real scenes and across robot data sources. Neither resource is itself a trained world model. Their relevance is the opportunity to learn relationships among observations, actions, and outcomes, subject to synchronization, sensing, and action conventions.

For a predictive-control comparison, we retain the action coordinate frame, command duration, control mode, observation frequency, and embodiment. Cartesian displacements, joint targets, and gripper commands are not interchangeable numerical vectors. A video of a human moving an object does not uniquely identify a command for a particular robot. Action alignment may require robot fine-tuning, inverse dynamics, or an embodiment-specific decoder. Table 2 separates resources by role; its extraction questions are not defects asserted for every resource.

2.2 Visual Prediction and Learned Simulation

Visual world models expose future observations that can be inspected, scored, or passed to another model. This is useful when an outcome is difficult to specify as a small state vector, but depicting a desired future differs from predicting the consequence of an action. UniPi generates a visual plan from a goal and extracts control actions [8]. UniSim learns interactive visual responses to commands and uses simulated experience for downstream learning [40]. One uses a future as an intermediate plan; the other uses a predictive environment as a source of interaction.

The distinction matters when an agent queries a model repeatedly. A coherent sequence can still misrepresent contact, a grasp, or the displacement induced by a command. We therefore separate conditioning signals from generated outputs. Text-conditioned motion, inferred latent actions, and actuator commands offer different control over a rollout. A single generated clip also differs from an interactive simulator whose next input depends on its own previous outputs.

Learned simulation connects data generation with policy improvement, but creates a validation dependency. A policy can exploit regularities of the learned environment. The relevant test is not only sample plausibility but whether a policy or evaluator trained with synthetic experience performs appropriately in an independently assessed environment. We treat generated experience as an additional data source whose utility must be measured, rather than an automatic substitute for physical interaction.

Interactive World Simulator provides a recent test of both uses [38]. Its action-conditioned latent dynamics support long autoregressive interaction, synthetic demonstration collection, and policy evaluation. On the reported tasks, some imitation policies trained on 100 generated demonstrations approach counterparts trained on 100 real demonstrations, while $\pi_{0.5}$ retains a substantial real-data advantage. Policy scores in the learned simulator are positively associated with physical scores over four tasks, but the comparison uses training-distribution initial states and shows a slight optimistic bias. This evidence supports a bounded surrogate role, not unrestricted replacement of physical evaluation.

2.3 Latent and Structured Predictive Representations

Latent models trade direct visual interpretability for representations shaped by learning objectives. Dreamer learns behavior through imagined latent trajectories [11]; TD-MPC2 uses a decoder-free latent model for trajectory optimization [13]. V-JEPA 2 separates action-free representation learning from action-conditioned adaptation for planning [3]. Thus, “latent” identifies a prediction space, not a complete training objective or control algorithm.

Structured models make selected physical entities explicit. Graph Network-based Simulators represent systems as interacting particles and learn their evolution through message passing [32]. Tesseract instead generates RGB, depth, and surface-normal sequences, reconstructs a time-varying point cloud, and uses an inverse-dynamics model to recover actions [44]. Its nine-task RL Bench study reports higher success than its video baseline on seven tasks, but does not include physical closed-loop control. This distinction matters: a geometrically structured future can aid action decoding without being a forward model queried by candidate robot actions.

Making geometry explicit does not remove state-estimation and control problems. A deployed system still needs to infer state from sensors, connect commands to boundary conditions, and account for unobserved quantities. Tesseract also identifies a concrete representation limit: its single-view RGB–depth–normal sequence captures only one visible surface.

Multimodal prediction raises a related issue. Vision, proprioception, touch, and force need not answer the same question. A contact event can be weakly visible but important for execution. OmniVTA predicts short-horizon visuotactile evolution, feeds predicted tactile features to a diffusion policy, and applies a 60-Hz controller to deviations between predicted and observed touch [45]. On six groups of xArm7 tasks, the full system exceeds its no-controller variant across the reported object, geometry, and perturbation settings. Its ablations support a role for tactile prediction in this system, while the single platform and sensor configuration limit wider conclusions.

We therefore ask what each modality makes observable, which future quantities are predicted, and how signals are synchronized. WorldArena 2.0 expands evaluation toward visuotactile modalities [34]; OmniVTA supplies a model-level case, but broader comparisons of force, contact, and deformable interaction remain necessary.

2.4 Generalization Across Scenes and Embodiments

Generalization should specify what changes between training and deployment. An unfamiliar object, a new goal, a shifted camera, and a different robot body impose different demands. Open X-Embodiment provides a data-centered view of heterogeneous robot learning [30]; DreamZero studies video–action modeling and transfer with different embodiments [41]. These directions motivate distinguishing transferable visual structure from the mapping to executable commands.

Our synthesis separates representation robustness, predictive validity, and action-interface compatibility. A system may share an encoder across robots while using separate action heads, or retain a predictor while recalibrating control.

Table 3. Prediction spaces and interfaces. A system may occupy several groups. The final column states a comparison question, not a ranking.

Prediction space	Representative sources	Information exposed	Comparison question
Compact or structured dynamics			
Latent trajectories	PlaNet; Dreamer; TD-MPC2 [11–13]	Features and task-related quantities	Which losses preserve decision-relevant information?
Predictive embeddings	V-JEPA 2 [3]	Feature targets and action adaptation	What changes between pretraining and planning?
Physical structure	GNS [32]	Particle states and interactions	How is state linked to sensing and actuation?
Visual and joint action interfaces			
Future images	UniPi; UniSim [8, 40]	Plans or conditioned rollouts	Is the future a target or an action consequence?
RGB–depth–normal	Tesseract [44]	Time-varying point clouds	Is geometry predicted from actions or decoded into actions?
Visuotactile futures	OmniVTA [45]	Contact evolution for policy and feedback	Which gain follows prediction rather than sensing alone?
Video and actions	DreamZero [41]	Joint video and action outputs	Which representation is executable?
Actions, state, value	Cosmos Policy [20]	Actions and outcomes for selection	When does prediction enter selection?

Both can be useful forms of transfer. Neither should be described as unrestricted embodiment generalization without recording which data, modules, and parameters changed during adaptation.

3 Integration with Robot Policies

3.1 Planning with Predicted Consequences

The most direct use of a world model is to compare candidate actions before executing one. PETS uses uncertainty-aware dynamics within trajectory sampling [6]; PlaNet and TD-MPC2 plan in learned latent spaces [12, 13]. Their predictive models enter an online decision procedure whose components include candidate generation, rollout, outcome scoring, and the feedback that replaces predictions with observations.

A generic receding-horizon decision can be expressed as

$$a_{t:t+H-1}^* = \arg \max_{a_{t:t+H-1}} \mathbb{E}_{p_\theta} \left[\sum_{k=0}^{H-1} \gamma^k r(y_{t+k+1}, a_{t+k}) \right], \quad (2)$$

with an initial action or chunk executed before replanning. Here r scores predicted outcomes and γ discounts future reward. This is a decision template, not an assertion that all cited systems implement the same optimizer. Learned values, terminal costs, constraints, and uncertainty penalties can change the objective. A better planner, evaluator, or candidate generator can also improve task success independently of the predictor.

Rollout length trades anticipation of delayed effects against reliance on increasingly uncertain predictions. Frequent feedback limits open-loop execution but cannot make an inaccurate model harmless: the first selected action can itself place a robot in a difficult state. Horizon, sampling budget, and control rate therefore belong in a method comparison

alongside architecture. A planning result is most interpretable when predictive quality and action-selection behavior are examined under the same task conditions.

3.2 Policy Learning with Imagined Experience

A world model can influence behavior without conducting a full online search at every control step. Dreamer learns an actor through imagined latent trajectories [11]. MBPO uses short synthetic rollouts branched from real experience for policy optimization [16]. These approaches move some predictive computation into learning. They differ from online planning in deployment cost, interaction requirements, and where model errors affect behavior.

DayDreamer anchors this discussion in physical sensing and actuation [39]. More generally, imagined learning involves two coupled feedback loops: interaction updates the model, and the model changes the policy that collects subsequent interaction. Analysis should record both loops, including the reward source and whether policy-induced distribution changes are reflected in model updates.

World-VLA-Loop makes the second loop explicit [26]. It trains an action-conditioned video model with successful and near-success trajectories, predicts both future frames and rewards, performs VLA reinforcement learning inside the model, and then uses rollouts from the updated physical policy to refine the next model. The reported two-task robot study improves after two iterations, but still requires task-specific demonstrations and new physical rollouts. The result therefore supports co-adaptation under policy shift, not elimination of real interaction.

Synthetic volume is not a sufficient measure of training value. MBPO’s restricted rollouts illustrate that model use can be selective rather than maximized [16]. For foundation-scale systems, the corresponding comparison should account for generating, filtering, and training on synthetic trajectories. Real interaction, synthetic interaction, and computation remain separate resources; a reduction in one is not automatically a reduction in all three.

3.3 Visual Plans and Joint World–Action Models

A generated future can serve as an intermediate action representation. UniPi synthesizes goal-conditioned video and extracts actions from it [8]. This separates choosing what should happen from translating that outcome into commands. It also identifies a failure boundary: a plausible plan can be unreachable by the current robot, and a correct plan can be poorly translated into motor actions. Plan quality and action grounding require separate examination.

Joint world–action models reduce this separation by learning future observations and commands together. DreamZero builds on a pretrained video diffusion model to generate video and actions [41]. Cosmos Policy encodes actions, future observations, and values within a video model’s latent generation process, and examines direct-policy and planning modes [20]. These examples illustrate the interface convergence discussed in recent surveys [14, 27], without implying that every video-based policy performs an explicit future rollout at execution time.

WAV combines a future-state model, a trajectory-value model, and action decoding through iterative latent inference [23]. Removing its latent planning stage reduces its reported LIBERO average from 98.1 to 96.4, with the largest drop on the Long suite. Its three-task physical comparison also favors WAV over a baseline that shares its video-pretrained model, although each task uses a separate model and only 15 trials. This is evidence for an implicit planning interface under those conditions, rather than a universal advantage over explicit search.

Table 4 compares the location of prediction in the pipeline. For a joint system, a predicted outcome may be consumed by action selection, shape training through an auxiliary loss, or remain another generated output. The mechanisms can coexist, but suggest different ablations. They should not be collapsed into an undifferentiated statement that a policy “uses a world model.”

Table 4. Prediction in policy learning and execution. Suggested comparisons are review criteria, not experiments asserted for every cited paper.

Integration route	Example	Model use	Controlled comparison
Online planning	PETS; TD-MPC2 [6, 13]	Candidate evaluation	Fix candidates and budget; vary predictor.
Imagined learning	Dreamer; MBPO [11, 16]	Policy/value training	Fix real-data budget; vary synthetic use.
Visual plan to action	UniPi [8]	Plan generation and decoding	Separate plan errors from decoder errors.
Joint generation	DreamZero [41]	Coupled video and actions	Vary predictive objective with matched training.
Joint model plus selection	Cosmos Policy [20]	Direct policy or planning	Direct versus planning mode; report compute.
Implicit latent planning	WAV [23]	Iterative future/value inference	Remove planning with the same model and data.
Co-evolving simulator and policy	World-VLA-Loop [26]	Model-based RL plus model refresh	Track physical data and gains per iteration.
Reasoner assistance	Astra studies [9, 35]	Proposal review or editing	Fix policy; vary observation and prediction inputs.

3.4 Reasoning-Policy Hybrids: The Astra Case

Recent Astra evaluations motivate distinguishing direct numerical action generation from a hybrid interface that reviews a learned policy’s proposals. In Su et al.’s technical report [35], Astra can correct actions proposed by $\pi_{0.5}$. On ten selected RoboDojo tasks with five aligned cases per task, the authors report 48% success for the hybrid and 26% for direct Astra control. The public-policy comparisons use published reference scores rather than same-seed reruns. These findings support a bounded discussion of complementary action-generation and correction roles; they do not isolate a future-prediction mechanism.

A separate evaluation by Zhang et al. [43] covers all 42 RoboDojo tasks, with 50 episodes per task for Astra. It reports 22.48% average success over 2,100 trials and a capability profile that varies substantially across task types. This broader result should be discussed alongside the selected-task hybrid study, while keeping their task sets and denominators separate.

The Galbot team’s September 29 preprint [9] extends the investigation across six domains, including cooperation with learned policies. Its gripper-manipulation discussion includes the same 48% RoboDojo hybrid result. We treat the overlapping result as part of one evidence series rather than a second independent replication. The paper also documents inference constraints, including paused physics during its locomotion reasoning calls. This makes the relation between simulated success and time-critical control an explicit part of the discussion.

Black Forest Labs’ FLUX 3 Action report [5] examines another combination: a fast action policy supplies proposals that Astra may accept, edit, replace, or stop. The report connects hybrid performance to inference cost and elapsed time, alongside task completion. This provides an industrial case for evaluating the entire decision loop, with the report’s own protocol and provenance retained.

Table 5. Interfaces and study scope in the Astra discussion. This is a provenance table, not a cross-paper performance leaderboard.

Source	Interface examined	Scope retained in the comparison
Su et al. [35]	Direct action generation; review of $\pi_{0.5}$ proposals	Selected tasks and aligned cases; public policy scores are reference results.
Zhang et al. [43]	LLM as policy	Full 42-task RoboDojo evaluation for Astra.
Galbot Team [9]	Direct and hybrid interfaces across domains	Domain-specific protocols; overlap with the earlier report.
Black Forest Labs [5]	FLUX 3 Action proposals with Astra review	Report-specific hybrid protocol and resource accounting.

For world-model research, these interfaces raise a specific attribution question. Does a hybrid improve because it predicts action consequences, identifies the correct target, repairs a proposal, or uses more computation? A useful comparison would hold the base policy, observations, control limits, and inference budget fixed while varying the information supplied to the reviewer: current observations, proposed actions, and explicit predicted consequences. This comparison is a proposed evaluation design of the survey, not an experiment established by the cited reports. It connects the Astra discussion to the article’s main question: when, and through which interface, does prediction change action selection?

4 Evaluation, Reliability, and Research Directions

4.1 Prediction Quality and Decision Utility

Evaluation should begin with the model’s claimed role. A representation learner needs evidence about relevant downstream features; an action-conditioned predictor needs tests of the relationship between commands and outcomes; a planner needs evidence about executed behavior. These are complementary questions, not a single evidence ladder that every dataset, theoretical result, and component must climb.

WorldArena compares perceptual quality with functional uses of embodied world models, including data generation, policy evaluation, and planning [33]. WorldArena 2.0 expands toward visuotactile modalities, interactive learning, and multiple simulated and physical platforms [34]. This broadens evaluation beyond generated-video appearance. The resulting protocols provide evidence about tested functions and settings, rather than universal certification of a high-scoring model.

Interactive World Simulator illustrates why the functional role also needs an external check [38]. Besides reporting video metrics over 192-step rollouts, it trains policies on generated demonstrations and compares policy rankings in the learned and physical environments. The observed positive association is more relevant to policy selection than pixel error alone, but it is not a calibration guarantee: the study covers four correlation tasks, training-distribution initial configurations, and a simulator that is slightly optimistic on three of them.

Policy benchmarks answer a related question. LIBERO studies knowledge transfer in robot learning [24]; CALVIN evaluates language-conditioned long-horizon manipulation [29]. Both can assess a complete system, but task success alone does not identify the responsible component. Jena et al. [17] foreground this distinction. Even within the same named suite, task subsets, demonstrations, initialization, and compute must be aligned before attributing differences to prediction.

Table 6. Evaluation targets and reporting requirements. These criteria match evidence to claims without assigning a universal evidence grade.

Target	Example setting	Informative observation	Protocol detail
Predictive content	Held-out trajectories	Error, consistency, or likelihood	Variable, horizon, units, and split.
Action sensitivity	Same history; varied commands	Predicted differences match observed consequences	Action semantics and intervention coverage.
Functional use	WorldArena series; Interactive World Simulator [33, 34, 38]	Utility as data source, evaluator, planner, or environment	Specific role and downstream system.
Task performance	LIBERO; CALVIN [24, 29]	Completion and sequential outcomes	Tasks, resets, trials, data, and budget.
Safety behavior	SafeDreamer; Recovery RL [15, 36]	Violations, interventions, and recovery	Cost, exposure, severity, and success.
Runtime feasibility	Physical closed-loop execution	Delay and valid control outputs	Hardware, sensing, action duration, and paused physics.

4.2 Uncertainty, Constraints, and Runtime Guardrails

Safety requires an account of what can go wrong, which quantity is constrained, and how the system responds. These choices lead to different claims. PETS propagates model uncertainty through candidate trajectories [6]; CPO optimizes reward under expected-cost constraints [1]; neither mechanism alone establishes that a physical robot will avoid every hazardous event. The uncertainty or cost must be tied to an action rule, a safety definition, and an operating distribution.

Formal certificates are conditional. Berkenkamp et al. combine a statistical dynamics model with Lyapunov verification and prove high-probability stability within an estimated region of attraction [4]. The result assumes, among other conditions, Lipschitz dynamics and policies, calibrated confidence bounds, a Lyapunov function, and an initial safe policy. Its experiment is a simulated inverted pendulum. Thomas et al. instead penalize trajectories that a short-horizon model predicts will become unsafe [37]. Their theorem requires a specified unsafe set, a finite horizon within which irrecoverable states fail, and a calibrated set-valued model; experiments on four MuJoCo tasks report fewer, not zero, violations.

Uncertainty-aware shielding offers a third interface. Yu et al. learn a distributional reachability certificate and a shield policy, reporting comparable return with fewer training violations on four simulated tracking and navigation tasks [42]. The paper is a 2022 preprint and lists physical-robot deployment as future work. SafeDreamer integrates cost-aware planning and Lagrangian updates with a learned world model [15]. Across Safety-Gymnasium simulation tasks it improves the reward–cost tradeoff, but its nearly zero-cost result is specific to the benchmark cost definition and does not certify open-world physical safety.

Recovery separates task progress from corrective behavior. Recovery RL learns task and recovery policies, using prior data to identify situations requiring intervention [36]. We include it as a complementary safety mechanism, not relabel it as a generative world model. Control barrier functions can restrict control through a specified safe-set condition [2]; their conclusions still depend on state estimation, dynamics, control authority, and feasibility.

At the system level, Kim et al. divide foundation-model-enabled robot safety into action, decision, and human-centered dimensions and propose monitoring and intervention layers [19]. This is an architectural position paper without an evaluation of the proposed stack. It therefore motivates separate decision and action gates, but does not

Table 7. Complementary safety mechanisms. Their scope follows their assumptions and evidence; rows are not ordered by safety strength.

Mechanism	Source	Function	Required check
Stability certificate	Berkenkamp et al. [4]	Certify a region of attraction	Initial safe set, confidence coverage, smoothness, and dimension.
Near-future avoidance	Thomas et al. [37]	Penalize predicted unsafe rollouts	Unsafe-set definition, model calibration, and sufficient horizon.
Reachability shield	Yu et al. [42]	Override actions using uncertainty-aware risk	Coverage, intervention rate, false alarms, and physical transfer.
WM cost planning	SafeDreamer [15]	Plan with learned costs and rewards	Cost-prediction error, exposure, and observed violations.
Recovery control	Recovery RL [36]	Switch to corrective behavior	Missed hazards, false alarms, and recovery success.
Modular guardrails	Kim et al. [19]	Separate monitoring and decision/action gates	Implemented modules, latency, independence, and incident reduction.
Predictive monitor	Liu et al. [25]	Request human help from latent rollouts	Trial denominator, supervisor variation, and intervention burden.
Semantic plan guard	RoboGuard [31]	Convert rules and a world model into plan constraints	Map integrity, rule coverage, adaptive attacks, and dynamics.
Verified safety module	FEARL [7]	Certify a small sensor-to-action module	Sensor calibration, verified domain, specification coverage, and transfer.

show that the architecture reduces physical incidents. Across these mechanisms, we distinguish guarantees under stated assumptions, empirical violation reduction, recovery demonstrations, and unvalidated design proposals.

Recent systems instantiate narrower parts of this architecture. Liu et al. use latent future rollouts to request human supervision before predicted manipulation failures [25]; two physical tasks improve over a matched-intervention baseline, although each task uses one human supervisor and the paper does not report a single physical-trial denominator. RoboGuard translates rules and a semantic world model into temporal-logic constraints for an LLM planner [31]. On a physical Jackal, it blocks all tested RoboPAIR attacks across 35 harmful prompts while preserving 10 safe behaviors, but assumes an accurate world model and does not cover all dynamics-dependent risks. FEARL separates a foundation-model controller from a small verifiable safety-action module [7]. In 18 physical navigation episodes, shielding changes the collision rate from 27.8% to zero with the same 61.1% task success; this evidence remains conditional on calibrated low-dimensional safety sensors and the verified input domain.

4.3 Operational Reliability and Attribution

Reliability depends on the complete observation–decision–execution loop. A planner can act on a stale observation; a controller can saturate; an action chunk can continue after the scene changes. This motivates reporting end-to-end latency and action timing, not just network inference time. The Astra evaluations in Section 3 also illustrate why

simulation timing must be explicit: pausing an environment removes consequences of reasoning delay that remain present during continuous physical operation.

Attribution requires controls at the relevant interface. To assess a predictive planner, hold candidate generation and scoring fixed where possible. To assess predictive pretraining, compare training signals with matched data and compute. For a reasoning-policy hybrid, distinguish additional observations, additional inference, proposal editing, and explicit consequence prediction. Not every study needs every comparison; the comparisons determine which explanation the study supports.

A practical reporting unit is an episode with its outcome, elapsed time, intervention history, and failure type. Aggregation should retain denominators and variation across tasks, seeds, or environments. Reports that reuse experiments should be linked as evidence families when appropriate. This is important for rapidly released systems where an early project report and an expanded preprint may contain overlapping trials.

4.4 Research Directions and Synthesis

From prediction scores to useful decisions. An important question is how changes in predictive quality alter decisions made with the model. This requires examining variables used by a planner, not just a visual aggregate score. Functional benchmarks provide one entry point [33]. Further comparisons should examine cases in which plausible predictions misrank actions or omit consequential alternatives.

A recent risk-informed world-model perspective sharpens this gap as three mismatches: likelihood versus risk, prediction versus intervention, and finite prediction horizon versus accumulated consequences [28]. Its proposed representation, counterfactual reasoning, memory, and assurance components form a research agenda rather than an evaluated architecture. The testable question is whether preserving rare, action-changing evidence alters action ranking, intervention, recovery, or abstention under matched information and compute.

Contact, hidden state, and embodiment. The next literature pass should connect structured and multimodal models to concrete sensing and actuation problems: contact prediction, deformable interaction, touch and force, and robots with different control conventions. Each comparison should distinguish observing a previously hidden variable, predicting it more accurately, and changing how a policy uses it. These are collection priorities, not claims that relevant literature is absent.

Adaptation and computation. Online updates, longer memory, and larger inference budgets change a system’s operating conditions. Comparisons should report adaptation data, regression, compute, and behavior under delayed or failed inference. Prediction and control need not run at the same rate. The design question is how to allocate computation while maintaining a response when prediction is uncertain, unavailable, or too slow.

Synthesis. World models connect experience to action through learned simulation, predictive representation, online planning, imagined learning, and joint world–action generation. These routes differ in what is predicted and where predictions are consumed. The review follows those dependencies from data and representation to behavior and reliability. The next stage is to broaden primary evidence for each route, especially contact-rich interaction, navigation and driving, and safety under distribution shift, while preserving the source-specific interpretation of each comparison.

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